Prodigy Anesthesia Quick Review

Bronchitis

What criteria is used to make a diagnosis of chronic bronchitis?	Chronic bronchitis is said to exist if a patient exhibits a productive cough on most days for three consecutive months for at least two years. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 534.*
What are the signs and symptoms of chronic bronchitis?	Frequent cough, copious sputum, elevated hematocrit, elevated PaCO2, and increased airway resistance. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 534-535.*
Is chronic bronchitis the same thing as chronic obstructive pulmonary disease (COPD)?	Chronic bronchitis is actually one of the two forms of COPD. Emphysema is the other form. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 361.*
How is chronic bronchitis related to cor pulmonale?	The chronic hypoxemia associated with severe chronic bronchitis will eventually lead to erythrocytosis, pulmonary hypertension, and right ventricular failure. Pulmonary hypertension and right ventricular failure are the hallmark findings in cor pulmonale. Only about 5% of patients with COPD will exhibit these symptoms. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 535.*
What are the most common factors that contribute to the development of bronchitis?	Cigarette smoking, exposure to air pollution, occupational exposure to dust, and pulmonary infections. Some patients may have a familial predisposition to the development of chronic bronchitis. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 534.*
How can chronic bronchitis be distinguished from emphysema?	Patients with chronic bronchitis typically exhibit frequent cough, copious sputum, an elevated hematocrit, increased markings (not hyperinflation) on chest x-ray, elevated PaCO2, normal elastic recoil, and increased airway resistance. In contrast, patients with emphysema have minimal sputum production, a less frequent cough, a normal hematocrit and PaCO2, decreased elastic recoil, hyperinflation on chest x-ray and a normal to slightly increased airway resistance. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 534-535.*

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What measures can be instituted preoperatively to reduce perioperative complications in patients with chronic bronchitis?	Antibiotics should be instituted for patients with purulent sputum and infiltrates on chest xray. Chest physiotherapy may be utilized to mobilize secretions. Adequate hydration should be ensured. The most effective way to reduce perioperative risk is to have the patient stop smoking at least 8 weeks prior to surgery. At 8 weeks of cessation, the pulmonary complication rate is the same as that for nonsmokers. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 361.*
What symptoms require postponing elective surgery in patients with chronic bronchitis?	Elective surgery should be postponed in these patients if they exhibit severe dyspnea, wheezing, pulmonary congestion, or a $PaCO_2$ greater than 50 mmHg. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 361.*

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